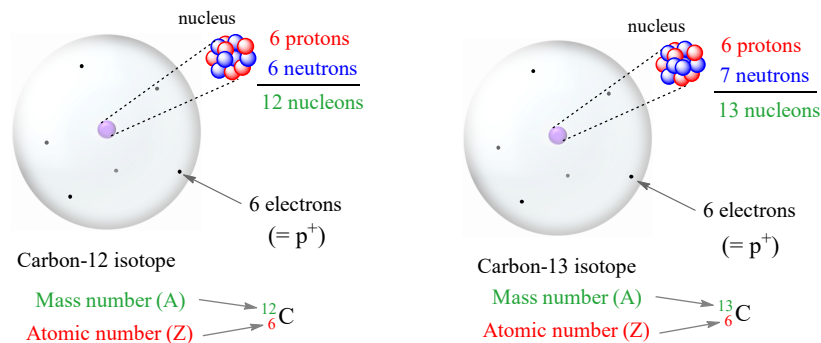


Atomic Structure

In a simplistic model, an atom is visualized as a sphere with electrons floating around the nucleus in the center. Below is a model for the atomic structure with the key terms and definitions shown for two carbon isotopes:



- Despite the extremely small size compared to the atom, nucleus makes most of the atom's mass.
- Most of the volume of the atom is empty space where the electrons float.
- Proton (p or p^+)** - positively charged subatomic particles with a positive electric charge of $1+$
- Neutron (n or n^0)** - neutral subatomic particles slightly heavier than proton
- Electron (e^-)** - a subatomic particle with a negative electric charge of $1-$
- Atomic number (Z)** - the number of protons in an atom. Always the same for the given element.
- Mass number (A)** - the sum of the number of neutrons and protons
- Nucleons** - a general term for both protons (p) and neutrons (n)
- Nuclide** - the nucleus of a given isotope
- Isotopes** - atoms with the same number of protons (atomic number) but different numbers of neutrons. Because of different number of neutrons, isotopes have different mass numbers.

The Periodic Table

Many elements show strong similarities to one another, and the periodic table arranges them in very useful patterns. Here are some of the most important ones you need to know:

- Elements with similar properties fall into the groups (vertical columns)
- Atomic number shows the number of protons in the element.
- Every atom has an equal number of electrons and protons, so atoms have no net electrical charge.
- Every element has one more proton than the previous one in the periodic table.

The elements are divided into three main categories - metals, nonmetals, and metalloids.

Metals are good conductors of heat and electricity while **nonmetals** are usually a poor conductor of heat and electricity. **Metalloids** have mixed properties of metals and nonmetals.

Metals lie on the lower left side and middle of the periodic table (yellow area) while nonmetals lie on the upper right side of the periodic table (green area). Metalloids lie along the zigzag diagonal line (blue area) that divides metals and nonmetals.

Remember the classification of the following groups:

Group 1A - Alkali metals: All very reactive with water to form products that are highly alkaline, or basic-hence the name alkali metals.

Group 2A - Alkaline earth metals: Less reactive than the metals in group 1A.

Group 7A - Halogens: Colorful, corrosive nonmetals found in nature only in form of salts. The group name halogen is taken from the Greek word *hals*, meaning "salt."

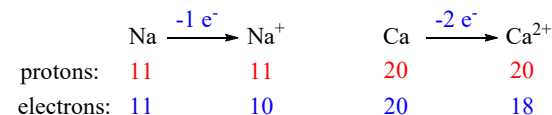
Group 8A - Noble Gases: Inert, unreactive gases. Og is a synthetic chemical element.

		Groups										Metals										Metalloids					Nonmetals					8A
		1A	2A																			3A	4A	5A	6A	7A						
Periods	1	H																												He		
	2	Li	Be																			5 B 10.81	C	N	O	F	Ne					
	3	Na	Mg																								Cl	Ar				
	4	K	Ca																								Br	Kr				
	5	Rb	Sr																								I	Xe				
	6	Cs	Ba																								Ft	Rn				
	7	Fr	Ra																								Ts	Og				

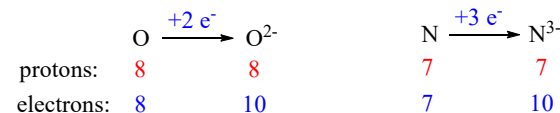
Ions

Ions are charged particles formed from atoms by gaining or losing an electron(s).

- Metals** tend to lose electron(s) and **become cations** (positively charged ions).



- Nonmetals** tend to gain electron(s) and **become anions** (negatively charged ions).



Notice that the number of protons is not changed, and the ions are charged because, **unlike atoms, their number of protons and electrons are not equal.**

This is important to keep in mind when determining the number of protons, neutrons, electrons, of atoms and ions. The following page is a worksheet on atomic structure and subatomic particles.

Use a periodic table to complete the following table. For missing symbols, add the specific isotope with the charge if needed.

	Symbol	Protons	Neutrons	Electrons	Charge	Mass no.
6	³⁵ Cl					35
5		11	12	10		
4	S ²⁻		16			
3				15	0	31
2		13			3+	27
1	Fe ²⁺					56
	a	b	c	d	e	f

The answers are on the right, and the explanations are below using the axes (a1, e5, etc.).

Solutions

- b1)** This is the atomic number of Fe. Looking up in the periodic table, it is 26.
- c1)** The number of neutrons varies for isotopes. Therefore, you need to subtract the number of protons from the atomic mass to determine the number of neutrons for this specific isotope: $56 - 26 = 30$
- d1)** For electrons, pay attention to the charge. 2+ means the Fe lost two electrons, and therefore, it has two electrons less than protons: $e = 26 - 2 = 24$
- e1)** The charge is shown on top-right corner of the symbol.
- a2)** First, you need to identify the element based on the atomic number (protons). It is 13, so we have Al. It shows a 3+ charge, so it is the aluminum ion, and not the atom.
- c2)** The number of neutrons varies for isotopes. Therefore, you need to subtract the number of protons from the atomic mass to determine the number of neutrons for this specific isotope: $27 - 13 = 14$.
- d2)** For electrons, pay attention to the charge. 3+ means the Al lost three electrons, and therefore, it has three electrons less than protons: $e = 13 - 3 = 10$.
- a3)** Normally, you'd look up the atomic number (protons) in the periodic table. However, it is not given here, so we need to determine the number of protons based on the number of electrons. Since there is no charge, these two must be equal. Therefore, there are 15 protons (based on d3) and we have phosphorous.
- b3)** From the periodic table, P has 15 protons. Again, we can also conclude this from the number of electrons since it is a neutral atom.
- c3)** The number of neutrons varies for isotopes. Therefore, you need to subtract the number of protons from the atomic mass to determine the number of neutrons for this specific isotope: $31 - 15 = 16$.
- b4)** This is the atomic number of sulfur. Looking up in the periodic table, it is 16.
- d4)** For electrons, pay attention to the charge. 2- means the S gained two electrons, and therefore, it has two electrons more than protons: $e = 16 + 2 = 18$.
- e4)** The charge is given in this example as well (2-).
- f4)** Mass number is the sum of the number of protons and neutrons: $16 + 16 = 32$.
- a5)** Like in (a3), we need to determine the number of protons based on the number of electrons. Since it has 1+ charge, there must be one more proton than electrons. So, it is $10 + 1 = 11$, and we have Na⁺.
- e5)** The charge is shown on the symbol: Na⁺, so it is 1+.
- f5)** Mass number is the sum of number of protons and neutrons: $11 + 12 = 23$.
- b6)** This is the atomic number of Cl which you find in the periodic table - 17.
- d6)** There is no charge, and therefore, the number of electrons is equal to the number of protons - 17.

This is the atomic number which you find in the periodic table.

Remember, $p + n = \text{atomic mass}$

For neutral atoms $p = e$

This is the difference of p's and e's.

$p + n = \text{atomic mass}$

Answers

	Symbol	Protons	Neutrons	Electrons	Charge	Mass no.
6	³⁵ Cl	17	18	17	0	35
5	Na ⁺	11	12	10	1+	23
4	S ²⁻	16	16	18	2-	32
3	P	15	16	15	0	31
2	Al ³⁺	13	14	10	3+	27
1	Fe ²⁺	26	30	24	2+	56
	a	b	c	d	e	f

Names of Ions

Ionic compounds are formed from cations and anions. For example, the Na^+ cations and Cl^- anions form the table salt (NaCl).

Most cations simply inherit the name of the metal they are derived from:

Li^+ = lithium ion Al^{3+} = aluminum ion Ca^{2+} = calcium ion

The ionic charge depends on the nature of the element. Remember the following ions in group 1A, 2A, and 7A as their charges are predictable and go based on their group:

1A	2A	Common Ions with Predictable Charges										3A	4A	5A	6A	7A	8A
H^+		<i>This is for nonmetals in monoatomic compounds.</i>												with metals - H^+			
Li^+	Be^{2+}													N^{3-}	O^{2-}	F^-	
Na^+	Mg^{2+}										Al^{3+}				S^{2-}	Cl^-	
K^+	Ca^{2+}	Sc^{3+}	Ti^{4+}	V^{2+} V^{3+}	Cr^{2+} Cr^{3+}	Mn^{4+}	Fe^{2+} Fe^{3+}	Co^{2+}	Ni^{2+}	Cu^+ Cu^{2+}	Zn^{2+}				Se^{2-}	Br^-	
Rb^+	Sr^{2+}								Pd^{2+}	Ag^+	Cd^{2+}					I^-	
Cs^+	Ba^{2+}																
Fr^+	Ra^{2+}																

Unlike group 1A and 2A metals, some transition metals can form cations with **different charges**. Therefore, their positive charge is indicated by a Roman numeral before the name of the metal.

Metals That Form Cations with Different Charges

Metal	Ion	Name	Ending
Chromium	Cr^{2+}	Chromium(II)	Chromous
	Cr^{3+}	Chromium(III)	Chromic
Iron	Fe^{2+}	Iron(II)	Ferrous
	Fe^{3+}	Iron(III)	Ferric
Cobalt	Co^{2+}	Cobalt(II)	Cobaltous
	Co^{3+}	Cobalt(III)	Cobaltic
Copper	Cu^+	Copper(I)	Cuprous
	Cu^{2+}	Copper(II)	Cupric
Tin	Sn^{2+}	Tin(II)	Stannous
	Sn^{4+}	Tin(IV)	Stannic
Mercury	Hg_2^{2+}	Mercury(I)	Mercurous
	Hg^{2+}	Mercury(II)	Mercuric
Lead	Pb^{2+}	Lead(II)	Plumbous
	Pb^{4+}	Lead(IV)	Plumbic

The ions of nonmetals in the periodic table make **binary compounds** which consist of *only two elements*. These ions are called **monoatomic ions**. There are also **polyatomic ions** which have certain rules for naming. However, at the end of the day, you are going to **memorize them**, so in this guide, we will only provide names of the most common polyatomic ions and a worksheet for additional practice.

Formulas and Names of Common Polyatomic Ions

Formula	Name
Singly charged anions	
CH_3CO_2^-	Acetate
CN^-	Cyanide
ClO^-	Hypochlorite
ClO_2^-	Chlorite
ClO_3^-	Chlorate
ClO_4^-	Perchlorate
H_2PO_4^-	Dihydrogen phosphate
HCO_3^-	Hydrogen carbonate (or bicarbonate)
HSO_4^-	Hydrogen sulfate (or bisulfate)
OH^-	Hydroxide
MnO_4^-	Permanganate
NO_2^-	Nitrite
NO_3^-	Nitrate
Doubly charged anions	
CO_3^{2-}	Carbonate
CrO_4^{2-}	Chromate
$\text{Cr}_2\text{O}_7^{2-}$	Dichromate
O_2^{2-}	Peroxide
HPO_4^{2-}	Hydrogen phosphate
SO_3^{2-}	Sulfite
SO_4^{2-}	Sulfate
$\text{S}_2\text{O}_3^{2-}$	Thiosulfate
Triply charged anion	
PO_4^{3-}	Phosphate

Ammonium (NH_4^+) is a common polyatomic cation.

Naming Ionic Compounds

To name an ionic compound, we list the name of the cation and then the anion. For **monoatomic anions**, the **ending** is changed to **"ide"**. This is not complicated, however, remembering the formulas and names of the ions is what takes some time and practice. Make sure to work on the practice problems on the last page. Some examples:

K_2S	$CaBr_2$	$AlCl_3$	MgO
Potassium sulfide	Calcium bromide	Aluminum chloride	Magnesium oxide

Examples of naming salts of **polyatomic ions**:

$BaSO_3$	Na_2CrO_4	$KClO_4$
Barium sulfite	Sodium chromate	Potassium perchlorate

Naming Molecular Compounds

A **molecule** is a group of two or more atoms held together by covalent bonds. Covalent bonds are usually formed between nonmetals. For example, Cl_2 , CH_4 , NH_3 , CO_2 .

When naming molecular compounds, the first element is treated as if it is a cation, so there is no change in its ending. The **second element** is named by adding **-ide** to the root of the element name.

- Prefixes are used to denote the numbers of atoms present.
- The prefix **mono-** is **never** used for naming the **first element**.
- Prefixes are not used for compounds containing a metal.

Prefixes - Number of Atoms	
mono-	1
di-	2
tri-	3
tetra-	4
penta-	5
hexa-	6
hepta-	7
octa-	8
nona-	9
deca-	10

Al_2S_3 - Aluminum sulfide, not dialuminum trisulfide.

SF_6	NCl_3	CO_2	N_2O_3
Sulfur Hexafluoride	Nitrogen trichloride	Carbon dioxide	Dinitrogen trioxide

- Molecular compounds containing hydrogen are also exceptions to using these prefixes.

B_2H_6	CH_4	SiH_4	NH_3	PH_3	H_2O	H_2S
diborane	methane	silane	ammonia	phosphine	water	hydrogen sulfide

Naming Hydrates

Hydrates contain a specific number of water molecules attached to them. The number of water molecules is represented with a dot and the Greek prefixes:

$MgSO_4 \cdot 7H_2O$ magnesium sulfate **heptahydrate** $CuSO_4 \cdot 5H_2O$ copper(II) sulfate **pentahydrate**

Naming Acids

For the purpose of learning the topic, recognize acids when you see H with any of the anions you learned so far: HCl , H_2SO_4 , HNO_3 , H_3PO_4 .

The name of the acid is derived from the anion found in it. These are the correlation patterns for names of monoatomic and polyatomic (oxoions) ions and their acids:

.....ide $\rightarrow$ hydro.....ic acid	ate $\rightarrow$ic acid	ite $\rightarrow$ous acid
Hypo.....ite $\rightarrow$ Hypo.....ous acid	Per.....ate $\rightarrow$ Per.....ic acid	

Names of Common Ions and Their Acids

Br^-	Bromide	HBr	Hydrobromic acid
CN^-	Cyanide	HCN	Hydrocyanic acid
NO_2^-	Nitrite	HNO_2	Nitrous acid
NO_3^-	Nitrate	HNO_3	Nitric acid
PO_4^{3-}	Phosphate	H_3PO_4	Phosphoric acid
SO_3^{2-}	Sulfite	H_2SO_3	Sulfurous acid
SO_4^{2-}	Sulfate	H_2SO_4	Sulfuric acid
ClO^-	Hypochlorite	$HClO$	Hypochlorous acid
ClO_2^-	Chlorite	$HClO_2$	Chlorous acid
ClO_3^-	Chlorate	$HClO_3$	Chloric acid
ClO_4^-	Perchlorate	$HClO_4$	Perchloric acid

Naming Bases

Most bases are substances that yield hydroxide ions (OH^-) when dissolved in water. To name a base, add the word hydroxide to the name of the cation:

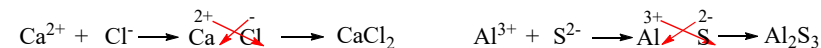
$NaOH$	$Ca(OH)_2$	$Fe(OH)_3$
Sodium hydroxide	Calcium hydroxide	Iron(III) hydroxide

Formula from the Name

There is one important thing to remember here: **In a neutral compound the sum of the ionic charges is equal to zero.** For molecular compounds too, the first atom is treated as a cation and the second as an anion, and the 'charge' is still zero:

Example: Write the formula of calcium chloride and aluminum sulfide.

For most formulas, you can place the ions next to each other, and cross down the charges as subscripts:



If there is a subscript, the polyatomic ion is put in parenthesis:

$Ca_3(PO_4)_2$	$Ba(NO_2)_2$
Calcium phosphate	Barium nitrite

Write the formula for each compound:

Nitrogen monoxide

Silver bromide

Sulfur dioxide

Lead(IV) iodide

Potassium carbonate

Hydrosulfuric acid

Sodium bicarbonate

Aluminum nitrate

Ferrous sulfide

Magnesium fluoride

Calcium cyanide

Manganese dioxide

Cuprous hydroxide

Silver chromate

Barium perchlorate

Cadmium sulfite

Potassium dichromate

Ammonium sulfate

Potassium permanganate

Lithium bisulfate

Zinc chlorate

Aluminium phosphate

Iron(III) oxide

Copper(I) nitrate

Mercuric hydroxide

Calcium carbide

Chromic sulfate

Stannous chloride

Ferric carbonate

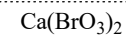
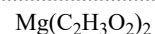
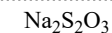
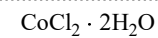
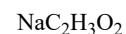
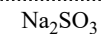
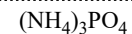
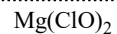
Ammonium hydroxide

Diphosphorus pentoxide

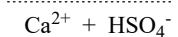
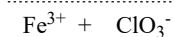
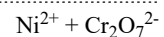
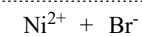
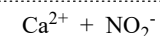
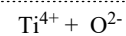
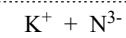
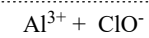
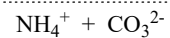
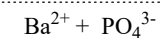
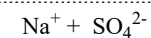
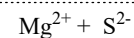
Carbon tetrachloride

Aluminum Oxide

Write an acceptable name for each compound:



Write the formulas of the compounds formed from the following ions:



Write the formula for each compound:

NO Nitrogen monoxide	MnO ₂ Manganese dioxide	Fe ₂ O ₃ Iron(III) oxide
AgBr Silver bromide	CuOH Cuprous hydroxide	CuNO ₃ Copper(I) nitrate
SO ₂ Sulfur dioxide	Ag ₂ CrO ₄ Silver chromate	Hg(OH) ₂ Mercuric hydroxide
PbI ₄ Lead(IV) iodide	Ba(ClO ₄) ₂ Barium perchlorate	CaC ₂ Calcium carbide
K ₂ CO ₃ Potassium carbonate	CdSO ₃ Cadmium sulfite	Cr ₂ (SO ₄) ₃ Chromic sulfate
H ₂ S Hydrosulfuric acid	K ₂ Cr ₂ O ₇ Potassium dichromate	SnCl ₂ Stannous chloride
NaHCO ₃ Sodium bicarbonate	(NH ₄) ₂ SO ₄ Ammonium sulfate	Fe ₂ (CO ₃) ₃ Ferric carbonate
Al(NO ₃) ₃ Aluminum nitrate	KMnO ₄ Potassium permanganate	NH ₄ OH Ammonium hydroxide
FeS Ferrous sulfide	LiHSO ₄ Lithium bisulfate	P ₂ O ₅ Diphosphorus pentoxide
MgF ₂ Magnesium fluoride	Zn(ClO ₃) ₂ Zinc chlorate	CCl ₄ Carbon tetrachloride
Ca(CN) ₂ Calcium Cyanide	AlPO ₄ Aluminium phosphate	Al ₂ O ₃ Aluminum Oxide

Write an acceptable name for each compound:

<i>Magnesium hypochlorite</i> Mg(ClO) ₂	<i>Ammonium phosphate</i> (NH ₄) ₃ PO ₄	<i>Sodium sulfite</i> Na ₂ SO ₃
<i>Dinitrogen trioxide</i> N ₂ O ₃	<i>Calcium chloride</i> CaCl ₂	<i>Potassium cyanide</i> KCN
<i>Sodium Acetate</i> NaC ₂ H ₃ O ₂	<i>Dinitrogen pentoxide</i> N ₂ O ₅	<i>Carbon monoxide</i> CO
<i>Barium sulfate</i> BaSO ₄	<i>Lead dioxide</i> PbO ₂	<i>Hydrogen peroxide</i> H ₂ O ₂
<i>Ammonia</i> NH ₃	<i>Cobalt(II) chloride dihydrate</i> CoCl ₂ · 2H ₂ O	<i>Potassium carbonate</i> K ₂ CO ₃
<i>Magnesium sulfite</i> MgSO ₃	<i>Copper(I) sulfide</i> Cu ₂ S	<i>Potassium perchlorate</i> KClO ₄
<i>Carbonic Acid</i> H ₂ CO ₃	<i>Dinitrogen monoxide</i> N ₂ O	<i>Potassium phosphate</i> K ₃ PO ₄
<i>Sodium peroxide</i> Na ₂ O ₂	<i>Titanium(IV) chloride</i> TiCl ₄	<i>Sodium thiosulfate</i> Na ₂ S ₂ O ₃
<i>Magnesium acetate</i> Mg(C ₂ H ₃ O ₂) ₂	<i>Magnesium nitride</i> Mg ₃ N ₂	<i>Tin(IV) chloride</i> SnCl ₄
<i>Calcium bromate</i> Ca(BrO ₃) ₂	<i>Diboron dichloride</i> B ₂ Cl ₂	<i>Carbon tetrabromide</i> CBr ₄

Write the formulas of the compounds formed from the following ions:

MgS Mg ²⁺ + S ²⁻	Na ₂ SO ₄ Na ⁺ + SO ₄ ²⁻	Ba ₃ (PO ₄) ₂ Ba ²⁺ + PO ₄ ³⁻
(NH ₄) ₂ CO ₃ NH ₄ ⁺ + CO ₃ ²⁻	Al(ClO ₃) ₃ Al ³⁺ + ClO ⁻	K ₃ N K ⁺ + N ³⁻
TiO ₂ Ti ⁴⁺ + O ²⁻	Ca(NO ₂) ₂ Ca ²⁺ + NO ₂ ⁻	NiBr ₂ Ni ²⁺ + Br ⁻
NiCr ₂ O ₇ Ni ²⁺ + Cr ₂ O ₇ ²⁻	Fe(ClO ₃) ₃ Fe ³⁺ + ClO ₃ ⁻	Ca(HSO ₄) ₂ Ca ²⁺ + HSO ₄ ⁻